



Continuous Stiffness Measurements and Nanoindentation Studies of bcc Mo: An Atomistic Approach

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Abstract Molecular dynamics (MD) simulations were performed to examine the continuous stiffness measurements (CSM) and nanoindentation behavior at 300 K temperature for body-centered cubic molybdenum (Mo). The nanomechanical behavior was examined by using a spherical indenter and the values of elastic modulus and hardness were extracted. In the nanoindentation model, we made use of a sinusoidal load in order to obtain the dynamical properties such as storage modulus, mechanical damping factor, and loss modulus. Open visualization tool (OVITO) was used to visualize the dislocation analysis, to track the indenter depth and grains response to the applied load. Dislocation loops were generated and found an increase in their areas as the indenter penetrated deep into the sample. This work provides useful insights toward a broad understanding of the structural, nano-mechanical, and damping properties of pure Mo.

Keywords Molecular dynamics · Spherical indenter · OVITO · Nanoindentation · Dynamic nanoindentation

1 Introduction

Mo is a bcc refractory metal, which can be used as a structural component in pressurized water reactors (PWRs) and boiling water reactors (BWRs) [1] due to its inviting thermal and mechanical characteristics. It is also a potential candidate in heat furnaces, glass melting components, powder semiconductor heat sinks, sputtering products in flat panel displays, and ADM for thermal sprays [2, 3]. The harsh irradiation in a reactor causes a large number of vacancies aggregating within the matrix which grow in three dimensions leading to cavities, which subsequently produce swelling. MD simulation of nanoindentation on glass performed earlier has shown that hardness is decreased after putting the material in a nuclear reactor environment [4]. Therefore, understanding the deformation behavior of Mo will be fruitful to optimize the design and functionality of cladding. MD simulation of nanoindentation is helpful in characterizing the deformation processes on atomic scale. In this work, we thoroughly studied the dynamic nanoindentation and nanoindentation behavior of Mo. The viscoelastic properties such as mechanical damping factor, loss modulus, and storage modulus are investigated. This study outputs the following results.

- E' the dynamic storage modulus,
- E'' the dynamic loss modulus,
- $\tan \delta$ or the mechanical damping factor

whereas $\tan \delta$ is the ratio of the loss modulus to the storage modulus and provides information regarding molecular mobility transition [2, 4]. The loss modulus and storage modulus represent the energy dissipated and stored during deformation in a cycle, respectively. Figure 1 represents the phasor diagram of the storage and loss modulus.

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